

CHARLOTTE-MONROE, NC, USA

WMO: 723194

Lat: 35.017N

Lon: 80.621W

Elev: 207

StdP: 98.86

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-3.9	-16.8	0.9	-2.3	-14.0	1.1	-0.2	9.9	9.9	8.8	9.6	2.1	30	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.9	23.4	33.6	23.3	32.4	23.0	25.2	31.4	24.7	30.5	24.1	29.7	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.9	27.3	23.0	18.2	26.7	22.6	17.7	26.3	77.8	31.3	75.5	30.6	73.5	29.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.3	DB	-9.8	36.7	2.2	1.7	-11.4	38.0	-12.7	38.9	-13.9	39.9	-15.5	41.1	(4)
(5)			WB	-11.0	26.2	2.0	0.9	-12.5	26.8	-13.6	27.3	-14.7	27.9	-16.2	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.8	7.6	11.6	16.5	20.7	25.0	26.5	25.9	22.8	16.8	11.0	7.7	(6)	
	DBStd	8.42	5.74	5.09	5.16	4.28	3.57	2.47	1.97	2.29	3.06	4.36	4.64	4.99	(7)	
	HDD10.0	459	157	100	44	4	0	0	0	0	0	4	42	107	(8)	
	HDD18.3	1652	389	303	216	84	21	1	0	0	5	81	222	331	(9)	
	CDD10.0	2844	26	32	93	199	333	449	512	492	385	213	73	36	(10)	
	CDD18.3	995	1	1	7	28	96	199	254	234	139	32	3	1	(11)	
	CDH23.3	8718	2	6	55	269	772	1818	2413	2083	1045	238	16	2	(12)	
	CDH26.7	3559	0	0	5	53	240	785	1125	918	381	52	1	0	(13)	
(14) Wind	WSAvg	2.6	2.9	3.1	3.1	3.1	2.6	2.3	2.2	2.0	2.4	2.4	2.6	2.6	(14)	
(15) Precipitation	PrecAvg	1117	88	76	102	90	81	106	116	103	99	86	79	91	(15)	
	PrecMax	1516	172	144	210	220	231	184	237	241	236	366	235	222	(16)	
	PrecMin	777	25	29	32	22	3	30	48	16	8	0	8	37	(17)	
	PrecStd	169	37	29	46	47	48	48	44	50	48	71	53	42	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.2	27.3	30.5	32.8	35.9	36.4	37.0	33.9	30.7	25.2	22.9	(19)	
		MCWB	16.3	17.2	17.0	19.2	22.0	23.2	24.2	23.2	22.0	22.0	17.2	17.4	(20)	
	2%	DB	19.4	21.0	24.9	28.1	31.0	33.9	35.0	34.2	32.3	27.8	22.9	20.3	(21)	
		MCWB	14.9	14.8	16.1	18.2	21.1	23.2	24.0	23.3	21.9	20.1	16.3	16.9	(22)	
	5%	DB	17.3	18.6	22.6	26.5	29.3	32.5	33.6	32.8	30.8	26.1	21.1	18.3	(23)	
		MCWB	13.6	13.8	15.0	17.4	20.7	22.8	23.8	23.5	21.9	19.0	16.1	15.5	(24)	
	10%	DB	15.0	16.3	20.3	24.6	27.7	31.1	32.3	31.4	28.9	23.9	19.0	16.3	(25)	
		MCWB	11.3	11.7	13.6	16.6	20.1	22.4	23.5	23.2	21.4	18.5	14.7	13.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	18.5	18.6	21.6	23.8	25.1	26.0	25.9	24.6	24.0	20.4	19.6	(27)	
		MCDB	20.7	21.7	24.1	27.4	29.0	31.9	32.8	31.3	29.7	27.1	22.1	21.0	(28)	
	2%	WB	16.3	16.8	17.5	20.2	22.6	24.3	25.3	25.2	23.8	22.5	18.7	17.7	(29)	
		MCDB	18.8	19.6	22.1	25.1	27.9	31.1	32.0	30.7	28.7	25.5	20.8	19.6	(30)	
	5%	WB	14.5	14.8	16.5	19.1	21.9	23.8	24.8	24.6	23.1	21.0	17.5	16.1	(31)	
		MCDB	16.9	17.9	21.1	23.7	27.0	30.2	31.1	29.8	27.7	24.0	19.7	17.9	(32)	
	10%	WB	12.0	12.3	15.2	17.9	21.1	23.2	24.3	24.0	22.4	19.6	15.7	14.0	(33)	
		MCDB	14.3	14.9	18.7	22.5	25.8	29.2	30.1	29.0	26.6	23.1	18.6	15.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.8	11.6	12.2	11.3	10.9	10.8	10.3	10.3	11.4	11.8	10.3	(35)	
		MCDBR	12.1	12.9	13.6	14.0	13.0	12.7	12.6	12.3	12.7	13.0	13.2	11.3	(36)	
	5% WB	MCWBR	8.4	8.1	6.7	6.2	5.2	4.2	3.8	3.7	4.3	5.9	7.8	7.9	(37)	
		MCWBR	10.4	11.4	11.6	11.6	10.9	11.2	11.0	10.2	10.1	9.5	9.9	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.312	0.322	0.358	0.403	0.460	0.486	0.539	0.515	0.431	0.381	0.330	0.310	(40)	
		taud	2.504	2.473	2.402	2.264	2.149	2.115	1.991	2.045	2.267	2.384	2.487	2.544	(41)	
	Ebn at Noon		895	925	913	882	833	808	762	773	827	844	865	877	(42)	
		Edh at Noon	83	95	111	134	153	158	178	166	126	103	84	76	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.66	3.31	4.38	5.49	6.08	6.44	6.04	5.51	4.69	3.86	2.93	2.30	(44)	
	RadStd		0.15	0.38	0.32	0.42	0.49	0.46	0.39	0.26	0.41	0.48	0.24	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+145	(46)	
(47)	Regional (71 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+165	+115	(47)	

Nomenclature: See separate page